Hypothesis Testing

Solutions

Starter

Solution 2

1) $X \sim B(20, 0.4)$. Calculate:

a) $P(X = 5)$

b) $P(X = 10)$

c) $P(X \leq 2)$

d) $P(X \geq 18)$

1a 0.075

1b 0.117

1c 0.0036

1d 0.00000504

2) Wanda rolls a fair die eight times.

a) Suggest a suitable model for the random variable X, the number of times the die lands on five.

2a $X \sim B\left(8, \frac{1}{6}\right)$

b) Calculate:

i. $P(X = 2)$

ii. $P(X \geq 4)$

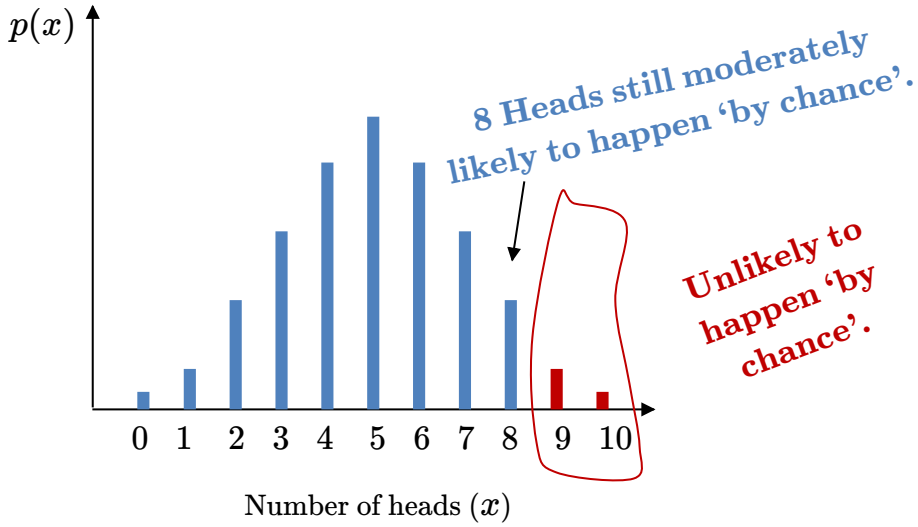
2b(i) 0.260

2b(ii) 0.0307



I throw a coin 10 times. For what numbers of heads might you conclude that the coin is biased towards heads? Why?

Our intuition is that the further away we are from the 'expected' number of heads (i.e. 5 heads out of 10), the more unlikely it is.





I throw a coin 10 times. For what numbers of heads might you conclude that the coin is biased towards heads? Why?

A hypothesis is a statement made about the value of a **population parameter** that we wish to test by collecting evidence in the form of a sample.

! The **null hypothesis**, H_0 is the default position, i.e. that nothing has changed, unless proven otherwise.

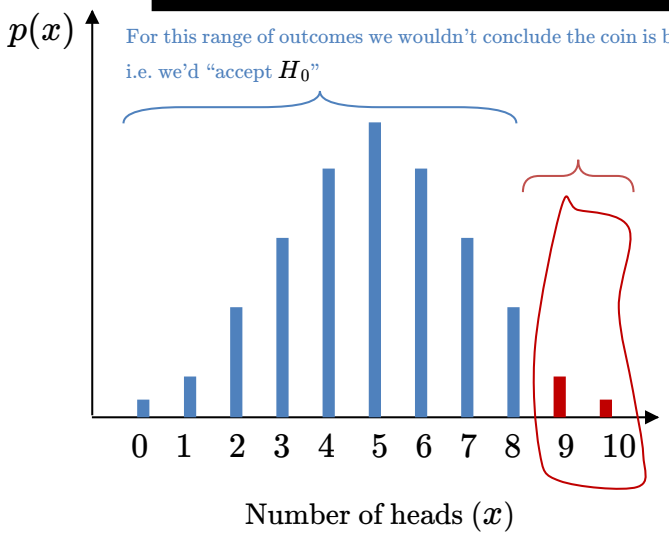
! The **alternative hypothesis**, H_1, is that there has been some change in the population parameter.

In this context...

We're asking "is the coin biased". This is making a statement about the probability p of getting Heads (i.e. the p in $B(n, p)$)

The 'default position' is that the coin is fair, i.e. $p = 0.5$.

The 'alternative' position is that the coin is biased towards heads, i.e. p is more than 0.5.



For this range of outcomes we'd conclude that this number of heads was too unlikely to happen by chance, and hence reject H_0 (i.e. that coin was fair) and accept H_1 (i.e. that coin was biased).



I throw a coin 10 times. For what numbers of heads might you conclude that the coin is biased towards heads? Why?

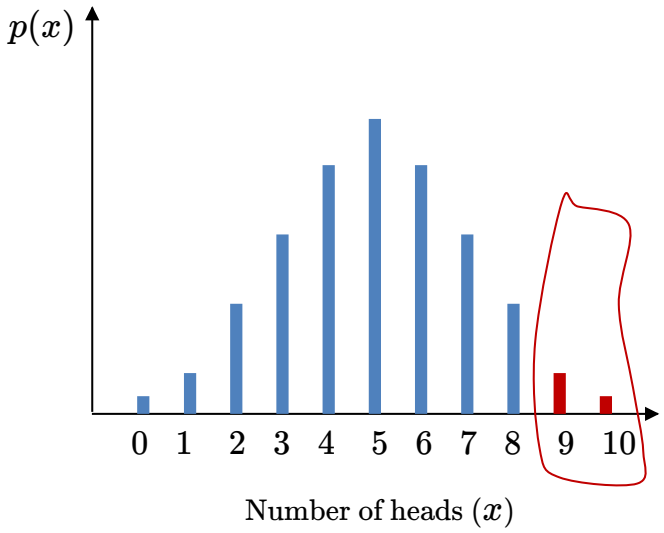
In this context...

In a hypothesis test, the evidence from the sample is a **test statistic**.

The test statistic X is **what we observed**, in this case, X is the number of heads seen in 10 throws.
Note that the test statistic **is a distribution** (i.e. across the possible things we might observe).

$$X \sim B(10, p)$$

noting that p is not known until we start making assumptions.



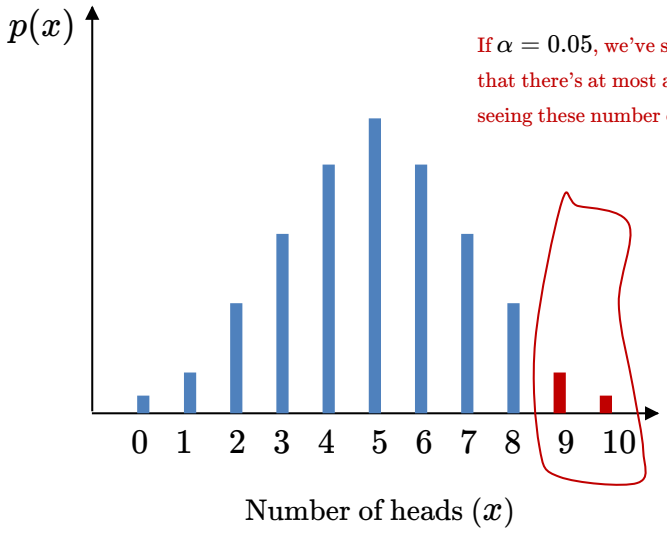


I throw a coin 10 times. For what numbers of heads might you conclude that the coin is biased towards heads? Why?

In this context...

! The **level of significance** α is the maximum probability where we would reject the null hypothesis.
This is usually 5% or 1%.

We said that if we saw a number of heads within ranges of outcomes that were sufficiently unlikely, then we'd rule out that the coin is fair and conclude it was in fact biased.
But how unlikely is 'sufficiently unlikely'? If $\alpha = 5\%$, then we'd find a region of outcomes where there's (at most) a 5% chance of one of these extreme values happening 'by chance' (i.e. if the coin was fair).



What is Hypothesis Testing? Solution 7

Hypothesis testing in a nutshell then is:

1. We have some hypothesis we wish to see if true (e.g. coin is biased towards heads), so. . .
2. We collect some sample data by throwing the coin (giving us our 'test statistic') and. . .
3. If that number of heads (or more) is sufficiently unlikely to have emerged 'just by chance', then we conclude that our (alternative) hypothesis is correct, i.e. the coin is biased.



Null Hypothesis and Alternative Hypothesis Solution 8

[Textbook] John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost.

We said that our two hypotheses are about the population parameter.

Suppose p is the probability of a coin landing heads.

Null hypothesis: $H_0 : p = 0.5$

Alternative hypothesis $H_1 : p > 0.5$

Under the **null hypothesis** H_0, we **assume that the population parameter is correct**, in this case, that it is a normal coin and the probability of Heads is 0.5

Under the **alternative hypothesis** H_1, there has been an underlying change in the population parameter, in this case that the coin is actually biased towards Heads.

The latter is known as a '**one-tailed test**' because we're saying the coin is biased one way or the other (i.e. $p > 0.5$ or $p < 0.5$).
But we could also have had the hypothesis 'the coin is biased (either way)', i.e. $p \neq 0.5$. This is known as a **two-tailed test**.

Further Example — Solution 9

[Textbook] An election candidate believes she has the support of 40% of the residents in a particular town. A researcher wants to test, at the 5% significance level, whether the candidate is over-estimating her support. The researcher asks 20 people whether they support the candidate or not. 3 people say they do.

a) Write down a suitable test statistic.

b) Write down two suitable hypotheses.

c) Explain the condition under which the null hypothesis would be rejected.

a

The number of people who say they support the candidate.

For a hypothesis test involving the binomial distribution, the test statistic is always the **count of successes**.

b

$H_0 : p = 0.4$

$H_1 : p < 0.4$

The alternative hypothesis is that the candidate is **overestimating** her support, so we're interested where **less than 40%** support them (more than 40% would not undermine the candidate's claim).

c

Null hypothesis would be rejected if the probability of 3 **or fewer people** supporting the candidate is less than 5%, given that $p = 0.4$

This is the hard bit!

We always calculate the probability of seeing this outcome **or more extreme** (in this case, 'more extreme' meaning even fewer the 3 people, because this takes us even further from the expected number of people out of the 20 (i.e. 8) who would support them.

The "$p = 0.4$" bit is because, as discussed before, we calculate the probability of seeing the observed outcome of 3 people (or more extreme) if it occurred **purely by chance** (the null hypothesis), i.e. if the candidate **did** have 40% support.

Test Your Understanding — Solution 10

In the UK, 5% of students turn up late to school each day. Miss Weir wishes to determine, to a 10% significance level if her school has a problem with attendance. She stands at the front gate one day and finds that 6 of the 40 students who pass her are late.

a) Write down a suitable test statistic.

b) Write down two suitable hypotheses.

c) Explain the condition under which the null hypothesis would be rejected.

a

The number of students who are late to school that day.

If there is indeed a problem with attendance (i.e. the school is not the norm) then we expect the proportion p of students late at Tiffin to be higher than 5%.

b

$H_0 : p = 0.05$

$H_1 : p > 0.05$

c

Null hypothesis would be rejected if the probability of 6 **or more** students being late to school is less than 10%, given that $p = 0.05$

Just to reiterate again what's going on here:

a) We assume that the school is the norm (i.e. 5% of students are late), and calculate the probability that 6 or more students would be late under this assumption.

b) If the probability of this happening by chance is sufficiently unlikely (less than the 10% significance level), we conclude that it probably isn't just by chance, and the school's lateness rate is worse than 5%.

Exercise 7A - Statistics 1 Pages 100-101 Solution 11

1. (a) Explain what a hypothesis test is. (b) Define a null and an alternative hypothesis and state their symbols. (c) Define a test statistic.
2. Classify each test as one- or two-tailed:

(a) $H_0 : p = 0.8, H_1 : p > 0.8$;

(b) $H_0 : p = 0.6, H_1 : p \neq 0.6$;

(c) $H_0 : p = 0.2, H_1 : p < 0.2$.

1a
A hypothesis is a statement about a population parameter. A test uses a sample or experiment to decide whether to reject that statement.

1b
H_0 is the assumption being tested. H_1 describes the alternative value or range of the parameter.

1c
A statistic calculated from the sample or experiment and used to test the hypothesis.

2a One-tailed (upper).

2b Two-tailed.

2c
One-tailed (lower).

Exercise 7A - Statistics 1 Pages 100-101 Solution 12

3. Dmitri tests whether a die is biased towards six. He rolls it 60 times, counting sixes.

(a) Describe the test statistic.

(b) Write a suitable null hypothesis.

(c) Write a suitable alternative hypothesis.

a
Number of sixes in 60 rolls.

b
$H_0 : p = \frac{1}{6}$, where p is the probability of a six.

c
$H_1 : p > \frac{1}{6}$.

Exercise 7A - Statistics 1 Pages 100-101 — Solution 13

4. Shell tests whether a coin is biased. She tosses it 100 times and counts heads. She says the test statistic is the probability of heads.
(a) Explain her mistake and give the correct statistic.
(b) Give a null hypothesis.
(c) Give an alternative hypothesis.

a

The probability of heads is the parameter. The statistic is the observed number of heads in 100 tosses.

b

$H_0 : p = \frac{1}{2}$.

c

$H_1 : p \neq \frac{1}{2}$.

Exercise 7A - Statistics 1 Pages 100-101 — Solution 14

5. Historically 10% of articles are faulty. In a sample of 100 from a new process, 8 are faulty. Test at the 5% level whether the proportion has fallen.
(a) Suggest a test statistic.
(b) State suitable hypotheses.
(c) Explain when the null hypothesis would be rejected.

a

Number of faulty articles among the 100.

b

$H_0 : p = 0.1,\ H_1 : p < 0.1$.

c

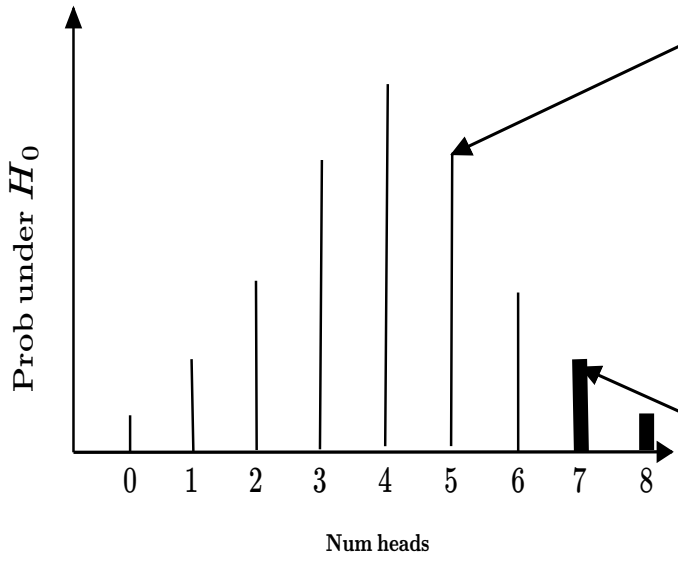
Reject H_0 if $P(X \leq 8)$ under H_0 is at most 0.05.

Critical Regions and Values

Solution 15

John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost. **What values would lead to John's hypothesis being rejected?**

As before, we're interested how likely a given outcome is likely to happen 'just by chance' under the null hypothesis (i.e. when the coin is not biased).



The probability of exactly five heads is only 22%. Why would seeing five heads be insufficient evidence of bias?

There are many possible outcomes. Five heads belongs to a range of outcomes collectively quite consistent with an unbiased coin.

However, there are values which collectively form a range of 'extreme values' where it would be unlikely that the coin would be unbiased. Their combined probability is limited by the level of significance set (e.g. 5%)

Critical Regions and Values

Solution 16

John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost. **What values would lead to John's hypothesis being rejected**, if the significance level was 5%?

What's the probability that we would see **6 heads**, or an **even more extreme value**? Is this sufficiently unlikely to support John's claim that the coin is biased?

$$\mathbf{P(X \geq 6) = 1 - P(X \leq 5)}$$
$$\mathbf{= 0.1445}$$

Insufficient evidence to reject null hypothesis (since $0.1445 > 0.05$).

What's the probability that we would see **7 heads**, or an **even more extreme value**?

$$\mathbf{P(X \geq 7) = 1 - P(X \leq 6)}$$
$$\mathbf{= 0.0352}$$

Since $0.0352 < 0.05$, this is very unlikely, so we reject the null hypothesis and accept the alternative hypothesis that the coin is biased.

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961

Critical Regions and Values

Solution 17

John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost. **What values would lead to John's hypothesis being rejected**, if the significance level was 5%?

✎ The **critical region** is the range of values of the test statistic that would lead to you rejecting H_0

If level of significance 5%, critical region?

We saw that 95% is exceeded when $X = 6$. This means $P(X \geq 7) = 1 - P(X \leq 6)$
$= 0.0352 < 5\%$
$\therefore 7 \leq X \leq 8$

✎ The value(s) on the boundary of the critical region are called **critical value(s)**.

Critical value:

7

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961
8	1

Tip: Use the first value AFTER the one in the table that exceeds 95%.

Quickfire Critical Regions

Solution 18

Determine the critical region when we throw a coin where we're trying to establish if there's the specified bias, given the specified number of throws, when the level of significance is 5%.

Coin thrown 5 times. Trying to establish if biased towards heads.

$p = 0.5, n = 5$

x	$P(X \leq x)$
0	0.0312
1	0.1875
2	0.5000
3	0.8125
4	0.9688

Critical region:

$X = 5$

Coin thrown 10 times. Trying to establish if biased towards heads.

$p = 0.5, n = 10$

x	$P(X \leq x)$
0	0.0010
1	0.0107
2	0.0547
...	...
7	0.9453
8	0.9893
9	0.9990

Critical region:

$9 \leq X \leq 10$

Coin thrown 10 times. Trying to establish if biased towards tails.

$p = 0.5, n = 10$

x	$P(X \leq x)$
0	0.0010
1	0.0107
2	0.0547
...	...
7	0.9453
8	0.9893
9	0.9990

Critical region:

$0 \leq X \leq 1$

Reminder: At the positive tail, use the value AFTER the first that exceeds 95% (100 - 5).

At the negative tail, we just use the first value that goes under the significance level.

Actual Significance Level

Solution 19

John wants to see whether a coin is unbiased or whether **it is biased towards coming down heads**. He tosses the coin 8 times and counts the number of times X, it lands head uppermost. **What values would lead to John's hypothesis being rejected**, if the significance level was 5%?

We saw earlier that the critical region was $X \geq 7$, i.e. the region in which John would reject the null hypothesis (and conclude the coin was biased).

We ensured that $P(X \geq 7)$ was less than the significance level of 5%.
But what actually is $P(X \geq 7)$?

$$P(X \geq 7) = 1 - P(X \leq 6) = 0.0352$$

This is known as the actual significance level, i.e. the probability that we're in the critical region. We expected this to be less than, but close to, 5%.

The **actual significance level** is the actual probability of being in the critical region.

C.D.F. Binomial table: $p = 0.5, n = 8$

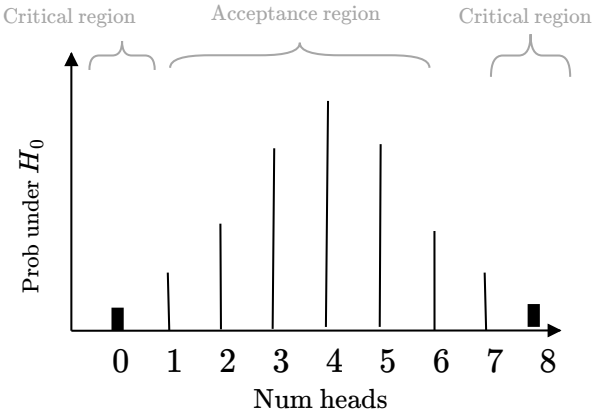
x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961
8	1

Two-tailed test

Solution 20

Suppose I threw a coin 8 times and was now interested in how may heads would suggest it was a **biased coin** (i.e. either way!). How do we work out the critical values now, with 5% significance?

We split the 5% so there's 2.5% at either tail, then proceed as normal:



Critical region at positive tail:

Look at closest value above 0.975 (then go one above):

X = 8

Critical region at negative tail:

Look at closest value below 0.025.

X = 0

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
...	...
6	0.9648
7	0.9961
8	1

Test Your Understanding

Solution 21

A random variable X has binomial distribution $B(40, p)$. A single observation is used to test $H_0 : p = 0.25$ against $H_1 : p \neq 0.25$.

a) Using the 2% level of significance, find the critical region of this test. The probability in each tail should be as close as possible to 0.01.

b) Write down the actual significance level of the test.

The $\neq$ indicates bias either way, i.e. two-tailed.

This means you find the closest to 0.01 (even if slightly above) rather than the closest under 0.01

a) Half of 0.02 is 0.01.

$P(X \leq 3) = 0.0047$

$0 \leq X \leq 3$

$P(X \geq 17) = 1 - P(X \leq 16) = 0.0116$

$17 \leq X \leq 40$

Critical region: $0 \leq X \leq 3$ or $17 \leq X \leq 40$.

b) $0.0047 + 0.0116 = 0.0163 = 1.63\%$

To ensure all method marks always show the probability of being in the critical region (even if you don't subsequently need the value!)

Note that X can't go below 0 or exceed 40.

C.D.F. Binomial table: $p = 0.25, n = 40$

x	$P(X \leq x)$
2	0.0010
3	0.0047
4	0.0160
5	0.0433
16	0.9884
17	0.9953
18	0.9983
19	0.9994

Warning: Textbook has several typos in this example.

Exercise 7B - Statistics 1 Pages 103-105

Solution 22

2. For $X \sim B(10, p)$, find the 5% critical region for $H_0 : p = 0.2$ against $H_1 : p > 0.2$.
3. A single observation from $B(20, p)$ is used to test $H_0 : p = 0.15$ against $H_1 : p < 0.15$. Find the 5% critical region.

2

Under H_0, $X \sim B(10, 0.2)$.

$P(X \geq 4) = 1 - 0.8791 = 0.1209 > 0.05$.

$P(X \geq 5) = 1 - 0.9672 = 0.0328 < 0.05$.

Critical value 5; region $X \geq 5$.

3

Under H_0, $X \sim B(20, 0.15)$.

$P(X \leq 1) = 0.1756 > 0.05$ but $P(X = 0) = 0.0388 < 0.05$.

Critical value 0; region $X = 0$.

Exercise 7B - Statistics 1 Pages 103-105 Solution 23

4. A single observation from $B(20, p)$ is used to test $H_0 : p = 0.4$ against $H_1 : p \neq 0.4$.
(a) Find the 5% critical region. (3 marks)
(b) Give the actual significance level. (1 mark)

a
Under H_0, $X \sim B(20, 0.4)$; allow 0.025 in each tail.
$P(X \leq 4) = 0.0510$, $P(X \leq 3) = 0.0160$.
$P(X \geq 13) = 1 - 0.9790 = 0.0210$; $P(X \geq 12) = 1 - 0.9435 = 0.0565$.
Critical region: $X \leq 3$ or $X \geq 13$.

b
$0.0160 + 0.0210 = 0.0370 = 3.70\%$.

Exercise 7B - Statistics 1 Pages 103-105 Solution 24

5. A test statistic has distribution $B(20, p)$. Given $H_0 : p = 0.18$ and $H_1 : p < 0.18$, find the critical region using a 1% level of significance.

5
Under H_0, $X \sim B(20, 0.18)$.
$P(X = 0) = 0.82^{20} = 0.0189 > 0.01$.
Even the smallest observation exceeds the 1% tail probability. The lower critical region is empty.

6. A random variable has distribution $B(10, p)$. A single observation is used to test $H_0 : p = 0.22$ against $H_1 : p \neq 0.22$.
(a) Using a 1% level of significance, find the critical region. The probability in each tail should be as close as possible to 0.005. (3 marks)
(b) Write down the actual significance level. (2 marks)

a
Under H_0, $X \sim B(10, 0.22)$.
$P(X = 0) = 0.0834$, so use an empty lower tail.
$P(X \geq 6) = 0.01039$ and $P(X \geq 7) = 0.001580$. The latter is closer to 0.005.
The critical region is $X \geq 7$.

b
Actual significance level: $0.001580 \times 100\% = 0.158\%$.

Exercise 7B - Statistics 1 Pages 103-105 — Solution 25

7. A component fails 3 times out of 10. An engineer tests 20 components made using a new system he thinks reduces failures.
(a) Describe the statistic. (1 mark)
(b) State hypotheses. (2 marks)
(c) Find the 5% critical region with probability as close to 0.05 as possible. (3 marks)
(d) Give the actual significance level. (1 mark)

a
Number of failed components.

b
$H_0 : p = 0.3,\ H_1 : p < 0.3.$

c
Under H_0, $X \sim B(20, 0.3)$.
$P(X \leq 2) = 0.0355$ and $P(X \leq 3) = 0.1071$.
The closer value is 0.0355. Region: $X \leq 2$.

d
$0.0355 = 3.55\%$.

Exercise 7B - Statistics 1 Pages 103-105 — Solution 26

8. An average of 12 seedlings in each tray of 36 survive. A gardener thinks a new fertiliser improves survival.
(a) Describe the statistic and give hypotheses. (3 marks)
(b) Find the 10% critical region. (3 marks)
(c) State the probability of incorrectly rejecting the null hypothesis. (1 mark)

a
X counts surviving seedlings. $H_0 : p = 1/3$, $H_1 : p > 1/3$.

b
Under H_0, $X \sim B(36, 1/3)$.
$P(X \geq 16) = 1 - 0.8906 = 0.1094 > 0.10$.
$P(X \geq 17) = 1 - 0.9416 = 0.0584 < 0.10$.
Region: $X \geq 17$.

c
$0.0584 = 5.84\%$.

Exercise 7B - Statistics 1 Pages 103-105 — Solution 27

9. One fifth of customers choose lasagne. A restaurant owner changes the recipe and thinks the proportion will change.

(a) Suggest a model and hypotheses. (3 marks)

She samples 25 customers.

(b) Find the 5% critical region. (4 marks)

(c) Give the probability of incorrectly rejecting the null hypothesis. (1 mark)

a

X counts customers choosing lasagne. $X \sim B(25, p)$; $H_0 : p = 0.2$, $H_1 : p \neq 0.2$.

b

Under H_0, $X \sim B(25, 0.2)$. Compare with 0.025 per tail.

$P(X = 0) = 0.0038$; $P(X \leq 1) = 0.0274$, closer to 0.025.

$P(X \geq 9) = 0.0468$; $P(X \geq 10) = 0.0173$, closer to 0.025.

Region: $X \leq 1$ or $X \geq 10$.

c

$0.0274 + 0.0173 = 0.0447 = 4.47\%$.

Doing a full one-tailed hypothesis test — Solution 28

Miss Weir tosses a coin 8 times and it comes up heads 6 times. She claims the coin is **biased towards heads**. With a significance level of 5%, test her claim.

X is number of heads.
p is probability of heads.
$X \sim B(8, p)$

STEP 1: Define test statistic X (stating its distribution), and the parameter p.

$H_0 : p = 0.5$
$H_1 : p > 0.5$

STEP 2: Write null and alternative hypotheses.

Assume H_0 is true, $X \sim B(8, 0.5)$
$P(X \geq 6) = 1 - P(X \leq 5)$
$= 1 - 0.8555$
$= 0.1445$

STEP 3: Determine probability of observed test statistic (or 'more extreme'), assuming null hypothesis.
i.e. Determine probability we'd see this outcome just by chance.

$14.45\% > 5\%$, so insufficient evidence to reject H_0.
There is insufficient evidence that the coin is biased towards heads.

STEP 4: Two-part conclusion:
1. Do we reject H_0 or not?
2. Put in context of original problem.

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961

NEW TO A LEVEL 2017: The probability of 'the observed value or more extreme' is known as the p-value.

Alternative method using critical regions

Solution 29

We can also find the critical region and see if the test statistic lies within it.

Miss Weir tosses a coin 8 times and it comes up heads 6 times. She claims the coin is **biased towards heads**. With a significance level of 5%, test her claim.

X is number of heads.
p is probability of heads.

$$X \sim B(8, p)$$

STEP 1: Define test statistic X (stating its distribution), and the parameter p.

$H_0 : p = 0.5$
$H_1 : p > 0.5$

STEP 2: Write null and alternative hypotheses.

$P(X \geq 7) = 1 - 0.9648 = 0.0352$
Critical region is $X \geq 7$

STEP 3 (Alternative): Determine critical region.

6 is not in critical region, so do not reject H_0.
There is insufficient evidence that the coin is biased towards heads.

STEP 4: Two-part conclusion:
1. Do we reject H_0 or not?
2. Put in context of original problem.

C.D.F. Binomial table: $p = 0.5, n = 8$

x	$P(X \leq x)$
0	0.0039
1	0.0352
2	0.1445
3	0.3633
4	0.6367
5	0.8555
6	0.9648
7	0.9961

More on p-values

Solution 30

Sheila wants to know if a coin is biased towards heads and throws it a large number of times, counting the number of heads. The p-value is less than 0.03.
Conduct a hypothesis test at the 5% significance level.

Let p be the probability of heads.
$H_0 : p = 0.5$
$H_1 : p > 0.5$
$0.03 < 0.05$ so reject H_0.
Sufficient evidence to suggest the coin is biased.

Ordinarily we'd calculate the probability of seeing the observed number of heads 'or more extreme'. But this has already been done for us (i.e. the p-value), so we just need to compare this against the threshold.

Further Example — Solution 31

[Textbook] The standard treatment for a particular disease has a $\frac{2}{5}$ probability of success. A certain doctor has undertaken research in this area and has produced a new drug which has been successful with 11 out of 20 patients. The doctor claims the new drug represents an improvement on the standard treatment. Test, at the 5% significance level, the claim made by the doctor.



X is number of patients for whom trial was successful.
p is probability of success in each patient.

$$X \sim B(20, p)$$

STEP 1: Define test statistic X (stating its distribution), and the parameter p.

$H_0 : p = 0.4$
$H_1 : p > 0.4$

STEP 2: Write null and alternative hypotheses.

Assume H_0 is true, so $X \sim B(20, 0.4)$

$$P(X \geq 11) = 1 - P(X \leq 10) = 0.1275$$

STEP 3: Determine probability of observed test statistic (or 'more extreme'), assuming null hypothesis.

$12.75\% > 5\%$ so not enough evidence to reject H_0.
There is insufficient evidence that the new drug improves the success rate.

STEP 4: Two-part conclusion:
1. Do we reject H_0 or not?
2. Put in context of original problem.

Test Your Understanding — Solution 32

Edexcel S2 Jan 2011 Q2

A student takes a multiple choice test. The test is made up of 10 questions each with 5 possible answers. The student gets 4 questions correct. Her teacher claims she was guessing the answers. Using a one tailed test, at the 5% level of significance, test whether or not there is evidence to reject the teacher's claim.
State your hypotheses clearly. **(6)**

$H_0 : p = 0.2,\ H_1 : p > 0.2$. Under H_0, $X \sim B(10, 0.2)$. (B1 B1)
$P(X \geq 4) = 1 - P(X \leq 3) = 1 - 0.8791 = 0.1209$. (M1 A1)
Alternatively $P(X \geq 5) = 0.0328$, giving the region $X \geq 5$
.
$0.1209 > 0.05$: insufficient evidence to reject H_0. The teacher's claim is supported. (M1 A1)

Note the mark for stating distribution of X under null hypothesis.

Note two-mark conclusion.

Example

Solution 33

1. Jessica complains that her die is biased so that she is less likely to get a six. She tests this at the 5% significance level.

(i) Write down her null and alternative hypotheses.

Jessica throws the die 20 times and gets just one six.

(ii) Find the p-value for this test.

(iii) What is Jessica's conclusion?

(i)

Let p be the probability of a six.

$H_0 : p = \frac{1}{6}, \quad H_1 : p < \frac{1}{6}.$

(ii)

Let X count sixes in 20 throws. Under H_0, $X \sim B(20, \frac{1}{6})$.

$P(X \leq 1) = (\frac{5}{6})^{20} + 20(\frac{1}{6})(\frac{5}{6})^{19} = 0.130$ (3 s.f.).

The p-value is 0.130.

(iii)

$0.130 > 0.05$, so do not reject H_0. There is insufficient evidence that the die is biased against a six.

Test your understanding

Solution 34

2. Hassan suspects his die is biased towards one. He tests at 10% and obtains 5 ones in 12 throws.

(i) Write down suitable hypotheses.

(ii) Find the p-value.

(iii) Give your conclusion.

(i)

$H_0 : p = \frac{1}{6}, \ H_1 : p > \frac{1}{6}.$

(ii)

Under H_0, $X \sim B(12, \frac{1}{6})$.

$P(X \geq 5) = 1 - P(X \leq 4) = 0.0364 < 0.1.$

(iii)

Reject H_0. There is evidence that the die is biased towards one.

Practice..

Solution 35

3. A bus company claims its service is on time in 90% of journeys. I suspect it is lower and test at 5%. In 15 journeys the bus is on time 10 times.
(i) Write down suitable hypotheses.
(ii) Find the p-value.
(iii) Give your conclusion.

(i)
$H_0 : p = 0.9,\ H_1 : p < 0.9.$

(ii)
Under H_0, $X \sim B(15, 0.9)$.
$P(X \leq 10) = 0.0127 < 0.05.$

(iii)
Reject H_0. There is evidence the service is on time in less than 90% of journeys.

.. and more practice.

Solution 36

4. A bus service was on time in 80% of journeys. The company claims improvements increased punctuality. I test at 10%. In 16 journeys the bus is on time 15 times.
(i) Write down suitable hypotheses.
(ii) Find the p-value.
(iii) Give your conclusion.

(i)
$H_0 : p = 0.8,\ H_1 : p > 0.8.$

(ii)
Under H_0, $X \sim B(16, 0.8)$.
$P(X \geq 15) = 1 - P(X \leq 14) = 0.1407 > 0.1.$

(iii)
Do not reject H_0. There is insufficient evidence that punctuality exceeds 80%.

Exercise 7C - Statistics 1 Pages 106-107

Solution 37

1. A single observation $x = 5$ is obtained from $B(10, p)$. At the 5% level test $H_0 : p = 0.25$ against $H_1 : p > 0.25$.
2. A single observation $x = 1$ is obtained from $B(10, p)$. At the 5% level test $H_0 : p = 0.4$ against $H_1 : p < 0.4$.

1

$H_0 : p = 0.25,\ H_1 : p > 0.25$. Under H_0, $X \sim B(10, 0.25)$.

$P(X \geq 5) = 1 - P(X \leq 4) = 0.0781 > 0.05$.

Do not reject H_0. There is insufficient evidence for $p > 0.25$.

2

$H_0 : p = 0.4,\ H_1 : p < 0.4$. Under H_0, $X \sim B(10, 0.4)$.

$P(X \leq 1) = 0.0464 < 0.05$.

Reject H_0. There is sufficient evidence for $p < 0.4$.

Exercise 7C - Statistics 1 Pages 106-107

Solution 38

3. A single observation $x = 10$ is obtained from $B(20, p)$. At the 5% level test $H_0 : p = 0.3$ against $H_1 : p > 0.3$.
4. A single observation $x = 3$ is obtained from $B(20, p)$. At the 1% level test $H_0 : p = 0.45$ against $H_1 : p < 0.45$.

3

$H_0 : p = 0.3,\ H_1 : p > 0.3$. Under H_0, $X \sim B(20, 0.3)$.

$P(X \geq 10) = 1 - P(X \leq 9) = 0.0480 < 0.05$.

Reject H_0. There is sufficient evidence for $p > 0.3$.

4

$H_0 : p = 0.45,\ H_1 : p < 0.45$. Under H_0, $X \sim B(20, 0.45)$.

$P(X \leq 3) = 0.0049 < 0.01$.

Reject H_0. There is sufficient evidence for $p < 0.45$.

Exercise 7C - Statistics 1 Pages 106-107

Solution 39

5. A single observation $x = 2$ is obtained from $B(20, p)$. At the 5% level test $H_0 : p = 0.28$ against $H_1 : p < 0.28$.
6. A single observation $x = 7$ is obtained from $B(8, p)$. At the 5% level test $H_0 : p = 0.32$ against $H_1 : p > 0.32$.

5

$H_0 : p = 0.28,\ H_1 : p < 0.28$. Under H_0, $X \sim B(20, 0.28)$.

$P(X \leq 2) = 0.0526 > 0.05$.

Do not reject H_0. There is insufficient evidence for $p < 0.28$.

6

$H_0 : p = 0.32,\ H_1 : p > 0.32$. Under H_0, $X \sim B(8, 0.32)$.

$P(X \geq 7) = 1 - P(X \leq 6) = 0.0020 < 0.05$.

Reject H_0. There is sufficient evidence for $p > 0.32$.

Exercise 7C - Statistics 1 Pages 106-107

Solution 40

7. A board-game die is suspected of giving too few sixes. One six appears in 12 rolls. Test at 5% whether the probability of a six is less than one sixth.

7

$H_0 : p = \frac{1}{6},\ H_1 : p < \frac{1}{6}$. Under H_0, $X \sim B(12, \frac{1}{6})$.

$P(X \leq 1) = 0.3813 > 0.05$.

Do not reject H_0. Insufficient evidence that a six is less likely than one sixth.

Exercise 7C - Statistics 1 Pages 106-107 — Solution 41

8. A skin treatment has a claimed success rate of 68%.

(a) Out of n patients, X successes occur. Give and justify a model.

Only 3 of 10 randomly sampled patients recover. It is thought the treatment is less effective than claimed.

(b) Test at 5%.

a

$X \sim B(n, 0.68)$: fixed n, two outcomes, independent patients and constant probability of success.

b

$H_0 : p = 0.68,\ H_1 : p < 0.68$. Under H_0, $X \sim B(10, 0.68)$.

$P(X \leq 3) = 0.0155 < 0.05$.

Reject H_0. There is evidence that the treatment is less effective than claimed.

Two-Tailed Tests — Solution 42

We have already seen that if we're interest in bias 'either way', we have two tails, and therefore have to split the critical region by **halving the significance level at each end**.

Over a long period of time it has been found that in Enrico's restaurant the ratio of non-veg to veg meals is 2 to 1. In Manuel's restaurant in a random sample of 10 people ordering meals, 1 ordered a vegetarian meal. Using a 5% level of significance, test whether or not the proportion of people eating veg meals in Manuel's restaurant is different to that in Enrico's restaurant.

Proportion eating veg meals at Enrico's is $\frac{1}{3}$

Let p be the proportion of people at Manuel's that order veg.

Let X be number of people eating veg meals.

$$\mathbf{H_0 : p = \frac{1}{3} \qquad H_1 : p \neq \frac{1}{3}}$$

If H_0 true then $X \sim B\left(10, \frac{1}{3}\right)$

$$\mathbf{P(X \leq 1) = P(X = 0) + P(X = 1)}$$
$$\mathbf{= 0.104 (3sf)}$$

$0.104 > 0.025$ therefore insufficient evidence to reject H_0.

There is no evidence that proportion of veg meals at Manuel's restaurant is different to Enrico's.

Half significance as 2 tailed.

Conclusion and what it means in context.

Test Your Understanding — Solution 43

Edexcel S2 Jan 2006 Q7a

A teacher thinks 20% of pupils in a school read the Deano comic regularly. He chooses 20 pupils at random and finds that 9 read the Deano.

(a)(i) Test at the 5% significance level whether the percentage differs from 20%. State your hypotheses clearly.

(ii) State all possible numbers of Deano readers from a sample of 20 that make the test significant at the 5% level. (9 marks)

(a)(i)

Let p be the population proportion. $H_0 : p = 0.2,\ H_1 : p \neq 0.2$. (B1 B1)

Under H_0, $X \sim B(20, 0.2)$. (B1)

$P(X \geq 9) = 1 - P(X \leq 8) = 1 - 0.9900 = 0.0100$. (M1)

$0.0100 < 0.025$, so reject H_0. (A1)

There is evidence that the proportion who read the Deano differs from 20%. (A1)

(ii)

$P(X = 0) = 0.8^{20} = 0.0115 < 0.025$; $P(X \leq 1) > 0.025$.

The significant outcomes are $X = 0$ or $9 \leq X \leq 20$. (B1 B1 B1)

Exercise 7D - Statistics 1 Pages 108-109 — Solution 44

1. A single observation $x = 10$ is obtained from $B(30, p)$. At the 5% level test $H_0 : p = 0.5$ against $H_1 : p \neq 0.5$.
2. A single observation $x = 10$ is obtained from $B(25, p)$. At the 10% level test $H_0 : p = 0.3$ against $H_1 : p \neq 0.3$.

1

$H_0 : p = 0.5,\ H_1 : p \neq 0.5$. Under H_0, $X \sim B(30, 0.5)$.

$E(X) = 15$, so use the lower tail and compare with 0.025.

$P(X \leq 10) = 0.0494 > 0.025$.

Do not reject H_0. There is insufficient evidence for $p \neq 0.5$.

2

$H_0 : p = 0.3,\ H_1 : p \neq 0.3$. Under H_0, $X \sim B(25, 0.3)$.

$E(X) = 7.5$, so use the upper tail and compare with 0.05.

$P(X \geq 10) = 1 - P(X \leq 9) = 0.1894 > 0.05$.

Do not reject H_0. There is insufficient evidence for $p \neq 0.3$.

Exercise 7D - Statistics 1 Pages 108-109

Solution 45

3. A single observation $x = 9$ is obtained from $B(10, p)$. At the 5% level test $H_0 : p = 0.75$ against $H_1 : p \neq 0.75$.
4. A single observation $x = 1$ is obtained from $B(20, p)$. At the 1% level test $H_0 : p = 0.6$ against $H_1 : p \neq 0.6$.

3

$H_0 : p = 0.75,\ H_1 : p \neq 0.75$. Under H_0, $X \sim B(10, 0.75)$.
$E(X) = 7.5$, so use the upper tail and compare with 0.025.
$P(X \geq 9) = 1 - P(X \leq 8) = 0.2440 > 0.025$.
Do not reject H_0. There is insufficient evidence for $p \neq 0.75$.

4

$H_0 : p = 0.6,\ H_1 : p \neq 0.6$. Under H_0, $X \sim B(20, 0.6)$.
$E(X) = 12$, so use the lower tail and compare with 0.005.
$P(X \leq 1) = 0.00000034 < 0.005$.
Reject H_0. There is sufficient evidence for $p \neq 0.6$.

Exercise 7D - Statistics 1 Pages 108-109

Solution 46

5. A single observation $x = 4$ is obtained from $B(50, p)$. At the 2% level test $H_0 : p = 0.02$ against $H_1 : p \neq 0.02$.
6. A coin lands heads 6 times in 20 tosses. Use a two-tailed 5% test to decide whether there is evidence of bias.

5

$H_0 : p = 0.02,\ H_1 : p \neq 0.02$. Under H_0, $X \sim B(50, 0.02)$.
$E(X) = 1$, so use the upper tail and compare with 0.01.
$P(X \geq 4) = 1 - P(X \leq 3) = 0.0178 > 0.01$.
Do not reject H_0. There is insufficient evidence for $p \neq 0.02$.

6

$H_0 : p = 0.5,\ H_1 : p \neq 0.5$. Under H_0, $X \sim B(20, 0.5)$.
$E(X) = 10$, so use the lower tail and compare with 0.025.
$P(X \leq 6) = 0.0577 > 0.025$.
Do not reject H_0. There is insufficient evidence that the coin is biased.

Exercise 7D - Statistics 1 Pages 108-109

Solution 47

7. Nationally 20% of patients experience complications after an operation. A hospital samples 20 records.
(a) Find critical regions for a two-tailed 5% test of whether its rate differs. Take each tail as close to 2.5% as possible. (5 marks)
(b) State the actual significance level. (1 mark)
(c) Eight patients had complications. Comment using the critical regions. (2 marks)

a

$H_0 : p = 0.2,\ H_1 : p \neq 0.2$. Under H_0, $X \sim B(20, 0.2)$.
$P(X \leq 1) = 0.0692$, $P(X = 0) = 0.0115$, closer to 0.025.
$P(X \geq 9) = 0.0100$, $P(X \geq 8) = 0.0321$, closer to 0.025.
Region: $X = 0$ or $X \geq 8$.

b

$0.0115 + 0.0321 = 0.0436 = 4.36\%$.

c

$X = 8$ is in the critical region. Reject H_0. The evidence suggests the hospital's rate differs from the national proportion.

Exercise 7D - Statistics 1 Pages 108-109

Solution 48

8. One in ten glass bowls has a hairline crack. After modifying the process, one of 20 sampled bowls is cracked. Test at 10% whether the proportion has changed. State your hypotheses. (7 marks)

8

$H_0 : p = 0.1,\ H_1 : p \neq 0.1$. Under H_0, $X \sim B(20, 0.1)$.
$E(X) = 2$, so use the lower tail and compare with 0.05.
$P(X \leq 1) = 0.3917 > 0.05$.
Do not reject H_0. Insufficient evidence that the proportion of cracked bowls has changed.

p-value

Doubling the calculated tail probability gives $2(0.391747) = 0.7835$.

Exercise 7D - Statistics 1 Pages 108-109

Solution 49

9. Agnetha's carrots have a 25% chance of being longer than 7 cm. After she changes fertiliser, 13 of 30 randomly sampled carrots exceed 7 cm. Test her claim that the probability has changed at 5%. State your hypotheses. (7 marks)

9

$H_0 : p = 0.25,\ H_1 : p \neq 0.25$. Under H_0, $X \sim B(30, 0.25)$.

$E(X) = 7.5$, so use the upper tail and compare with 0.025.

$P(X \geq 13) = 1 - P(X \leq 12) = 0.0216 < 0.025$.

Reject H_0. There is evidence that the probability of exceeding 7 cm has changed.

Exercise 7D - Statistics 1 Pages 108-109

Solution 50

10. A standard blood test correctly diagnoses a disease with probability 0.96. A cheaper test is claimed to have the same probability. In a clinical trial it diagnoses 63 of 75 patients correctly. Test the claim at 10%, stating your hypotheses. (7 marks)

10

$H_0 : p = 0.96,\ H_1 : p \neq 0.96$. Under H_0, $X \sim B(75, 0.96)$.

$E(X) = 72$, so use the lower tail and compare with 0.05.

$P(X \leq 63) = 0.00004174 < 0.05$.

Reject H_0. There is evidence that the new test has a different probability of success.